



Adverse effects of pristine and aged polystyrene microplastics in mice and their Nrf2-mediated defense mechanisms with tissue specificity

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Abstract

Health hazards associated with microplastics (MPs) remain largely unknown, and the effects of aged MPs, one of their persistent forms, are poorly characterized. Male ICR mice were intratracheally instilled with 0.01 and 1 mg/day pristine and ultraviolet (UV)-aged polystyrene microplastics (PS and APS) with an average diameter of 4–5 µm daily for 1 week. UV irradiation caused the PS to have a rough surface, become fragmented, and increase their carbonyl groups. Both PS and APS caused structural damage to the mouse gut, liver, spleen, and testis. Inflammatory infiltration in liver, swollen and congested gut, and loose spleen globules, as well as the loose interstitium of the seminiferous tubules in testis were found in 1 mg/day APS group. Increases in serum alanine aminotransferase and immunoglobulin A levels in 1 mg/day APS group ($p < 0.05$) demonstrated that APS exposure could induce greater liver and spleen functional damage than PS. Meanwhile, triglyceride and total cholesterol levels in liver were enhanced in 1 mg/day APS group ($p < 0.05$). Superoxide dismutase and glutathione contents in 0.01 and 1 mg/day APS groups significantly decreased ($p < 0.05$), which suggesting that PS and APS could interfere with the antioxidant capacity in mice. Nuclear factor erythroid 2-related factor 2 (Nrf2) and heme oxygenase-1 (HO-1) levels in the PS and APS groups showed significant increases in the liver and testis ($p < 0.05$), and a significant decrease in the spleen ($p < 0.05$), which were analyzed to get a first survey for Nrf2/HO-1-mediated tissue-specific defense mechanisms. In conclusion, acute exposure to PS and APS induced potential metabolic disorders, and APS could produce more serious immune damage and reproductive toxicity. These findings provide new insights in health risk assessment of aged MPs.

Keywords Aged polystyrene microplastics · Tissue damage · Nrf2/HO-1 · Immune response · Liver dysfunction · Mouse

Introduction

Plastic particles with a diameter of < 5 mm, known as microplastics (MPs), are increasingly regarded as a global environmental threat and human health hazard (Jiang et al. 2020). MPs have been found in sea salts at a concentration of 0–1674 particles/kg, and the abundance of MPs ranges from 5 ± 2 to 91 ± 14 particles/L in drinking water, with an average of 18 ± 7 particles/L (Kim et al. 2018; Shruti et al. 2020). Numerous studies have demonstrated that MPs can accumulate in the tissues such as the liver and gut of various

marine organisms and be transported upwards through the food chain (Wong et al. 2020; Dela-Torre 2020). Moreover, humans can be directly exposed to MPs through seafood and drinking water (Zhang et al. 2020; Nicole 2021). The health risks of human exposure to MPs are gradually being recognized as a global problem. Studies indicate that each person consumes approximately 3×10^4 – 6×10^4 MP particles per year, and MPs have been found in human placenta tissue (Cox et al. 2019; Ragusa et al. 2021). Therefore, it can be concluded that the widespread usage and persistence of MPs have led to their accumulation in the organisms, inducing a greater exposure risk to human health.

MP toxicology research suggests that MPs could be accidentally ingested by aquatic organisms (Jeong and Choi 2019). Exposure to polystyrene microplastics (PS-MPs) induces developmental delays and reduces fecundity in *Paracyclops nana* in a dose-dependent manner (Jeong et al. 2017). PS can accumulate in the liver of juvenile *Eri-ocheir sinensis* and promote oxidative stress (Yu et al. 2018).

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In brine shrimp, chronic exposure to PS at environmentally relevant concentrations disturbs signaling pathways (Suman et al. 2020). Moreover, PS can cause liver lipid metabolism disorders, microbiota dysbiosis, and gut inflammation in zebrafish (Lu et al. 2016; Wan et al. 2019). However, there are no sufficient data on the health effects and toxicity of MPs in mammals. Studies have found that 0.1 mg/day PS can cause oxidative stress and induce energy and lipid metabolism abnormalities in mouse liver, as well as potentially toxic effects such as neurotoxicity (Deng et al. 2017; Wang et al. 2022). PS (1000 µg/L) also induces gut microbiota disorders and disrupts the gut barrier and metabolic function in mice (Lu et al. 2018; Jin et al. 2019). These results indicate that MPs produce toxic effects on mammals. However, the effects and mechanisms of acute exposure to MPs on various organs in mammals remain to be studied.

The release of MPs into different environments alters their physical and chemical properties, such as specific surface area, electric charge, and adsorption capacity, through ultraviolet (UV) irradiation, hydrolysis, and temperature. MPs with changed physical and chemical properties become aged MPs (Jahnke et al. 2017; Liu et al. 2019a, b). Previous studies have focused on pristine MPs, while a few studies have shown that the aging of MPs promotes their ingestion by marine zooplankton (Hu and Palić 2020; Vroom et al. 2017). Aged MPs can inhibit the growth of *Chlorella vulgaris* through oxidative stress (Fu et al. 2019). In addition, nanoplastics (NPs) are formed during the degradation and fragmentation of plastic materials under various environmental conditions (Lambert and Wagner 2016). PS particles sized 100 nm and 5 µm cause different cytotoxic and inhibit efflux pump effects in human Caco-2 cells (Wu et al. 2019). Furthermore, charged PS-NPs more easily accumulate in organisms than pristine MPs (Walczak et al. 2015a, 2015b). Sun et al. (2020) found that differentially charged NPs show distinct patterns of accumulation in plant *Arabidopsis thaliana*, producing ecological effects and impacting agricultural food safety and sustainability. Humans can be exposed to MPs directly through food such as vegetables and seafood (Oliveri Conti et al. 2020; Leung et al. 2021). In summary, aging of MPs not only alters their physical and chemical properties, but also has different toxic effects, inducing potential health risks to humans. Therefore, it is necessary to further assess the environmental and health risks of MPs.

Mice are often used as experimental animal models because they are easy to breed, with high reproduction rate and high genetic similarity to humans, which is often used to assess the potential health threats to humans caused by environmental pollutants (Hofker 2003). The nuclear erythroid 2-related factor 2/heme oxygenase-1 (Nrf2/HO-1) axis plays a key role in antioxidant mechanisms (Ayıközü et al. 2021), and recent research demonstrates that PS activates the Nrf2/HO-1 pathway, inducing potential reproductive toxicity in

mice (Hou et al. 2021). However, it is not clear whether APS exposure will damage mouse organs through the Nrf2/HO-1 pathway. Therefore, the aim of this study was to elucidate the differences in the toxicity of pristine and aged PS-MPs to different organs in mice and to screen out the sensitivity indicators of the potential health risks of MPs in aged states.

Materials and methods

Experimental materials

PS-MPs with a particle size of 5.24 ± 1.15 µm were purchased from Tianjin BaseLine ChromTech Research Centre. The specific aging process of PS-MPs was performed according to the method of Wang et al. (2021), specifically: 50 mg PS-MPs was weighed and placed in a quartz sand test tube, and 10 mL ultrapure water was added to each test tube at a speed of 260 rpm/min. PS-MPs were continuously exposed to UV light for 7 days in a 500 W high-pressure mercury lamp chemical reaction instrument (Xujiang Electromechanical Plant, Nanjing, China). After 7 days of UV irradiation, PS-MPs were repeatedly washed 2–3 times with absolute ethanol (GR), then filtered through a 1 µm nylon net filter with a vacuum pump, dried in an oven at 40°C, and counted under a stereomicroscope. Scanning electron microscopy (SEM; Quanta-250FEG, FEI, USA), transmission electron microscopy (TEM; H-7650, Hitachi, Japan), Fourier transform infrared spectroscopy (FTIR; Nicolet-iN10MX, Thermo, USA), X-ray photoelectron spectroscopy (XPS, PHI 5000 VersaProbe, ULVAC-PHI, Japan), and a particle size analyzer (ZEN1600, Malvern Instruments, England) were used to characterize the pristine/aged PS.

Animals and exposure test

Male ICR mice were selected as experimental exposure subjects because of their high sensitivity and superior physical health indicators. Specific pathogen-free (SPF) male ICR mice (6 weeks; 24 ± 2 g) were purchased from SPF Biotechnology Co. Ltd. (Beijing, China). The mice were reared in the SPF-level experimental animal center of Nanjing University (Jiangsu, China). Mice were provided random access to tap water and food and maintained at a temperature of 20.4–23.6 °C and humidity of 40–60% under a 12-h light/dark cycle. The animal experiment was approved by the Institutional Animal Care and Use Committee of Nanjing University (Nanjing, China). Experimental ICR mice were randomly divided into five groups of seven individuals: (1) control group (Control); (2) 0.01 mg/day PS group (low-dose pristine PS, LP-PS); (3) 1 mg/day PS group (high-dose pristine PS, HP-PS); (4) 0.01 mg/day aged PS group (low-dose aged PS, LA-PS); (5) 1 mg/day aged PS group (high-dose

aged PS, HA-PS). Mice were exposed to PS and APS by gavage every day for 1 week. Experimental concentrations were based on reports on the intake of MPs in humans and existing experimental exposure concentrations (Cox et al. 2019; Deng et al. 2020).

Biochemical detection

Mouse serum samples were collected from heart blood by centrifugation (3000 rpm at 25 °C for 10 min) and stored at − 80 °C for subsequent experiments. All biochemical detection in serum was measured by assay kits according to the manufacturer's instructions (Nanjing Jiancheng Bioengineering Institute, China). Immunoglobulin A (IgA) concentrations were measured to assess immunity, and the sensitivity of the IgA enzyme-linked immunosorbent (ELISA) kit (NO. H108-1) was 0.05–15 mg/mL. Triglyceride (TG) (NO. A11-1-1) and total cholesterol (TC) (NO. A111-1-1) levels were detected to measure lipid accumulation, and the sensitivities of the TG and T-CHO kits were 0–9.04 mmol/L and 0–10.34 mmol/L, respectively. Lactate dehydrogenase (LDH, A020-2-2), alanine aminotransferase (ALT/GPT, NO. C009-2-1), and aspartate aminotransferase (AST/GOT, C010-2-1) were measured to assess liver function. Superoxide dismutase (SOD) (NO. A001-3) and glutathione (GSH) (NO. A006-2-1) contents were detected to reflect oxidative stress levels.

Histological assay

Mouse colon, ileum, liver, kidney, spleen, testis, brain, lung, and heart were removed for histopathological analysis at the end of the experiment. Mouse organs were then rinsed with phosphate buffered saline (PBS) and fixed in 10% formalin solution overnight. The tissue samples were embedded in paraffin, sliced, and dewaxed with xylene. The 4-μm slices were then stained with hematoxylin and eosin (H&E) and scanned using a microscope (Nikon Eclipse E100, DS-U3, Japan). Tissue images were captured using Case Viewer (3DHISTECH, Hungary).

Immunofluorescence

The paraffin fixation method was the same as above (H&E staining). Paraffin slices of liver, spleen, and testis tissues were dewaxed, rehydrated, and repaired. The autofluorescence of dried slices was quenched with a fluorescence quencher, and bovine serum albumin (BSA) was used to seal the slices. The slices were then incubated with anti-Nrf2 (GB13148, 1: 100, Service Bio, Wuhan, China) monoclonal antibodies. The dried tissues were incubated with a secondary antibody at room temperature. After being decolorized and washed, 4,6-diamidino-2-phenylindole (DAPI) was used

to restrain the nucleus (DAPI, G1012, Service Bio, Wuhan, China). The decolorized and washed slices were sealed with an anti-fluorescence quencher (anti-fluorescence quencher, G1401, Service Bio, Wuhan, China). Finally, the slices were scanned with a microscope (Nikon Eclipse C1, DS-U3, Japan), and the images were captured using Case Viewer and analyzed with Image-Pro Plus 6.0 (Media Cybernetics, USA).

Immunohistochemistry

The paraffin fixation, slicing, dewaxing, rehydration, and repair process of liver, spleen, and testis were the same as immunofluorescence. Three percent hydrogen peroxide solution was used to block endogenous peroxidase in slices, and 3% BSA was used to seal slices with serum. The slices were then incubated with anti-HO-1 (GB11845, 1: 1000, Service Bio, Wuhan, China) monoclonal antibodies. At room temperature, the slices were incubated with a secondary antibody (horseradish Peroxidase-labeled). After being decolorized and washed, 3,3'-diaminobenzidine (DAB) was used to restrain the nucleus (DAB, G1211, Service Bio, Wuhan, China). The nucleus was counterstained with hematoxylin, dehydrated, and mounted. Finally, the slices were scanned with a microscope (Nikon E100, DS-U3, Japan), and the images were captured using Case Viewer and analyzed with Image-Pro Plus 6.0.

Statistical analyses

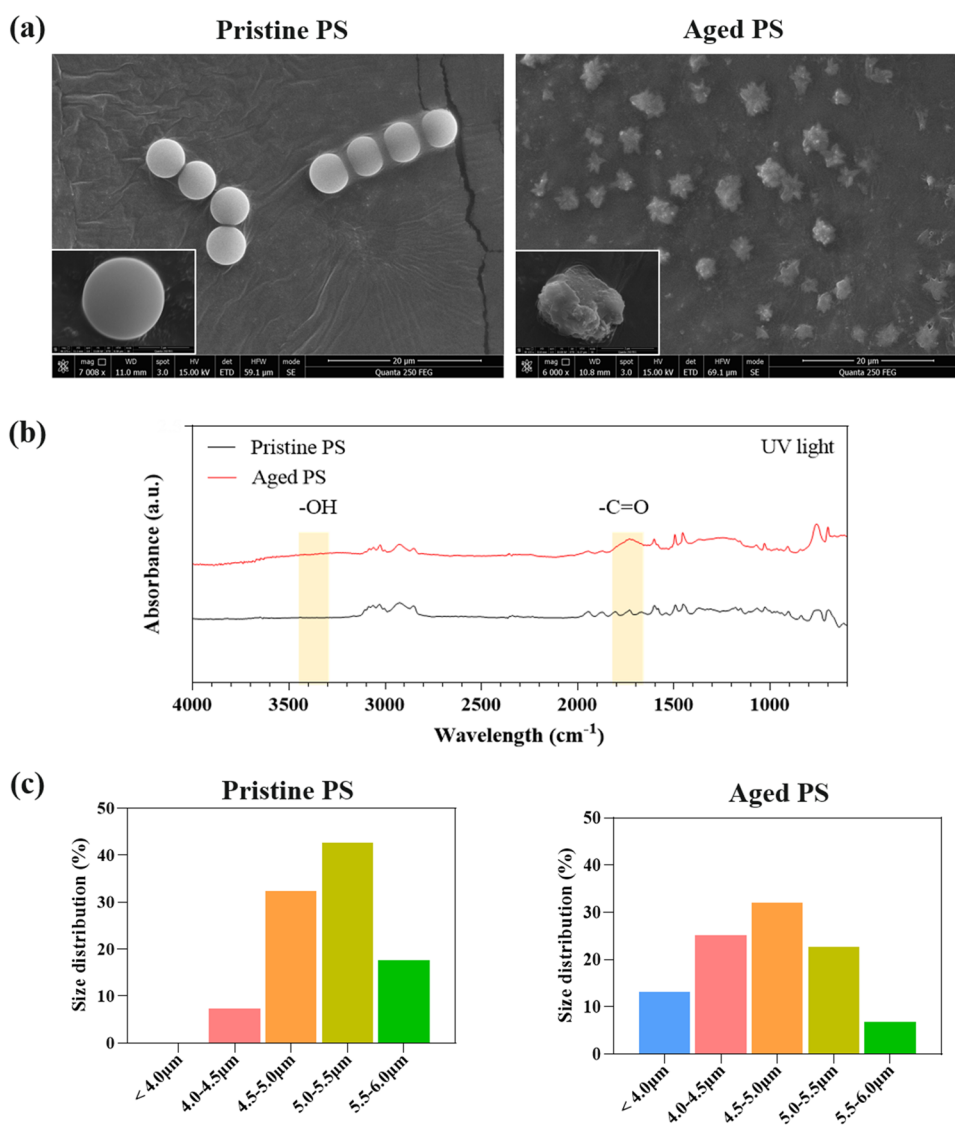
Statistical analyses and graphs were performed using SPSS 13.0 and GraphPad Prism 8.0 (San Diego, CA, USA) (Jeong et al. 2017; Wang et al. 2022). According to the sample distribution (Shapiro–Wilk normality test), the differences among different groups were assessed by one-way analysis of variance (ANOVA) with Tukey's post hoc test (Piñeiro-Hermida et al. 2022). The data were shown as mean ± standard errors; $p < 0.05$ was considered the presence of a statistically significant, and $p < 0.01$ was considered statistically extremely significant.

Results and discussion

Characterization of PS and APS

The characterization and particle size distribution of PS-MPs were obtained (Fig. 1a and c). As shown in Fig. 1a, SEM images of the PS-MPs showed that the wrinkles and cracks appeared on surfaces of the aged PS-MPs during the aging process. FTIR spectra showed that, after UV irradiation treatments, changed peaks were found at 3419.1 cm^{−1}

Fig. 1 Characterization of pristine/aged PS-MPs. **a** SEM images. **b** FTIR images. **c** Particle size distribution

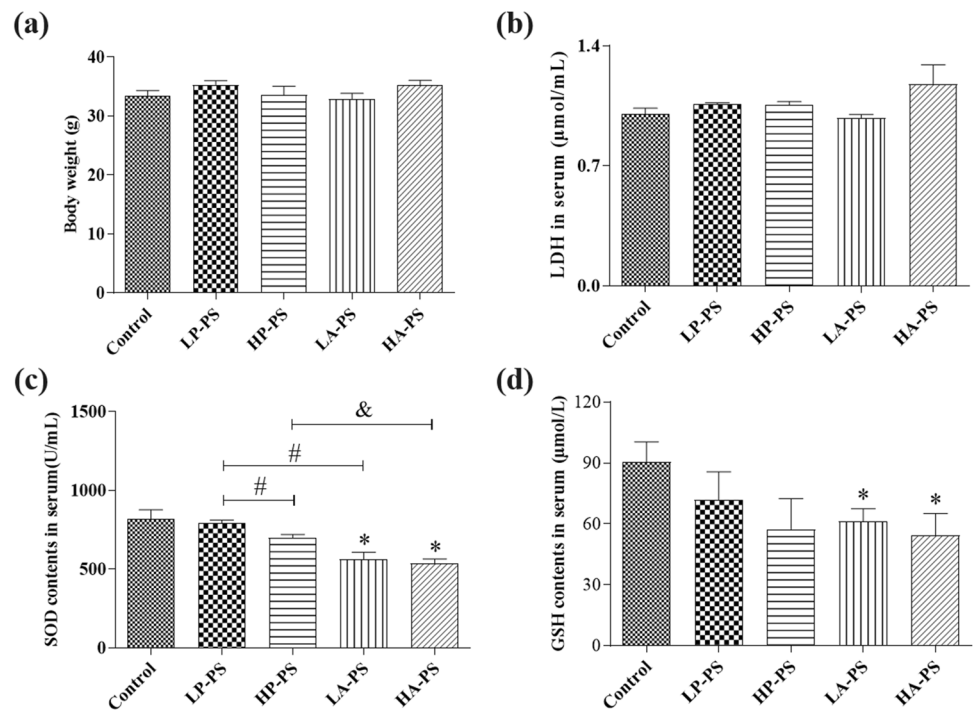


and 1731.7 cm⁻¹ for APS (Fig. 1b), and these peaks were assigned to increases in hydroxyl (-OH) and carbonyl (-C=O) group telescopic vibration (Fig. S1). A total of 42.63% and 32.37% of pristine PS were 5.0–5.5 µm and 4.5–5.0 µm, respectively, with 5.0–5.5 µm being the dominant size. After 168 h of UV irradiation, 31.96% of APS was 4.5–5.0 µm, and 25.20% and 13.14% of APS were 4.0–4.5 µm and < 4.0 µm, respectively, suggesting that PS-MPs splintered into pieces during the aging process (Fig. 1c). These are similar to previous characterization and analysis results of APS (Liu et al. 2019a, b; Song et al. 2018; Jin et al. 1992). Carbonyl-containing unsaturated chemicals can induce higher toxicity, which is attributed to the hydrogen bonding with carbonyl groups (Schultz et al. 2005). Thus, it is crucial to study the toxicity differences between pristine and aged MPs in mice.

PS and APS exposure decreased SOD and GSH contents in mice

There were no significant changes in body weight and LDH in PS and APS exposure groups compared with the control group (Fig. 2a and b). The most frequently mentioned endpoints induced by MPs are those related to oxidative stress, including antioxidant and other signaling pathways (Espinoza et al. 2018; Xie et al. 2020; Jeong and Choi 2019). SOD and GSH can effectively remove the oxygen free radicals generated in the process of cell metabolism and are important indexes to evaluate whether the antioxidant capacity is damaged (Pigeolet et al. 1990; Suzuki et al. 1999). Therefore, it was necessary that SOD and GSH levels were determined to assess differences in toxicity between pristine and aged MPs. Compared with the control group, SOD levels showed a decreasing trend in the LP-PS and HP-PS groups

Fig. 2 Effects of PS and APS exposure on the body weight and enzyme activity of mouse serum. **a** Body weight. **b** LDH. **c** SOD. **d** GSH. * $p < 0.05$, compared with the control group. # $p < 0.05$, compared with the LP-PS group. & $p < 0.05$, compared with the HP-PS group



(Fig. 2c), and they significantly decreased in the LA-PS and HA-PS groups ($p < 0.05$). In addition, SOD levels in the APS groups were significantly decreased compared with those in the PS groups. Similarly, GSH levels also showed a decreasing trend in the PS groups (Fig. 2d), and significantly decreased in the LA-PS and HA-PS groups ($p < 0.05$). Liang et al. (2021) showed that PS exposure can significantly inhibit the activities of CAT, SOD and other antioxidant enzymes in rotifer *Brachionus calyciflorus*, impairing its antioxidant capacity, growth, and reproduction. PS-MP exposure also resulted in a significant decrease in SOD of larval zebrafish (Yang et al. 2020). Thus, decreases in SOD and GSH levels suggest that exposure to APS could significantly disrupt the antioxidant system in mice, indicating that antioxidant capacity could be used as a biomarker to assess the health risks of aged MPs.

PS and APS disturbed the Nrf2/HO-1-induced antioxidant system in mice

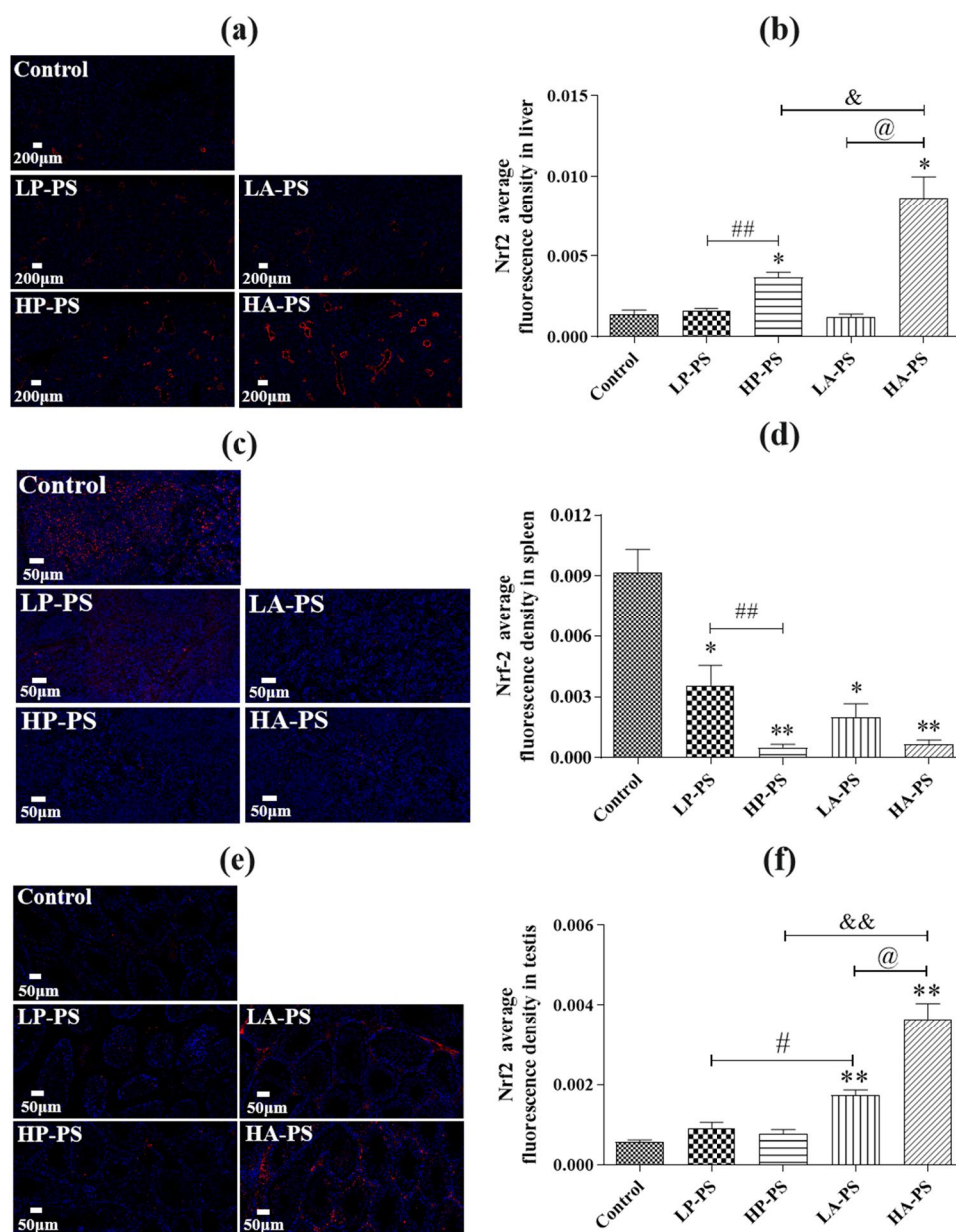
Nrf2 protein is expressed in various organs, such as the heart, liver, kidney, spleen, and testis, and plays a key role in antioxidant mechanisms (Niture et al. 2014; Moi et al. 1994; Tang et al. 2014). Based on the decreases in SOD and GSH levels induced by exposure to pristine and aged PS-MPs, the levels of Nrf2 in the liver, spleen, and testis were detected by immunofluorescence. As shown in Fig. 3a and b, the expression levels of Nrf2 in liver in the HP-PS group were significantly higher than those in the control group ($p < 0.05$). PS promotes the expression of Nrf2, inducing

defense mechanisms, reducing the reproductive capacity of *Paracyclopina nana* (Jeong et al. 2017), and activates the Nrf2/HO-1/nuclear factor kappa-B pathway, inducing potential reproductive toxicity in mice (Hou et al. 2021). Compared with the HP-PS group, exposure to HA-PS not only enhanced the expression of Nrf2 in liver ($p < 0.01$), but also significantly increased the Nrf2 levels in testis (Fig. 3b and c, $p < 0.01$), illustrating that aged PS-MPs incurred more serious oxidative damage in the mouse liver and testis by promoting Nrf2 expression. Correspondingly, aged MPs can aggravate growth inhibition of *Chlorella vulgaris* through oxidative stress (Fu et al. 2019).

The Nrf2/HO-1 pathway plays an important role in antioxidant process, and HO-1 is closely related to antioxidation, anti-inflammatory and anti-apoptotic processes (Ayıkğöz et al. 2021). In this study, PS and APS were found to induce the pathological damage in liver, spleen, and testis due to activate the Nrf2/HO-1 pathway in mice. From the analyzed data of Fig. 4, it clearly showed that PS and APS exposure significantly promoted HO-1 expression in liver and testis (Fig. 4a, b, e, and f, $p < 0.05$) and decreased HO-1 level in spleen (Fig. 4c and d, $p < 0.05$). In both liver and spleen, the levels of HO-1 showed a significant difference in LA-PS and HA-PS groups compared to LP-PS and HP-PS, respectively ($p < 0.05$), which revealed that APS could induce more potential tissue damage than PS. The resulting Nrf2/HO-1 signal path changes will be useful in understanding the mechanisms in the PS and APS-induced oxidative stress.

Different concentrations of PS and APS resulted in variations of the expression levels of Nrf2 in mouse tissues, while

Fig. 3 Effects of PS and APS exposure on Nrf2 levels in mouse organs. **a** Immunofluorescence to detect Nrf2 level in liver. **b** Nrf2 average fluorescence density in liver. **c** Immunofluorescence to detect Nrf2 level in spleen. **d** Nrf2 average fluorescence density in spleen. **e** Immunofluorescence to detect Nrf2 level in testis. **f** Nrf2 average fluorescence density in testis. * $p < 0.05$, ** $p < 0.01$, compared with the control group. # $p < 0.05$, ## $p < 0.01$, compared with the LP-PS group. & $p < 0.05$, && $p < 0.01$, compared with the HP-PS group. @ $p < 0.01$, compared with the LA-PS group. For **a**, **c**, and **e**, the nucleus stained with 4', 6-diamidino-2-phenylindole was blue under ultraviolet excitation, and the positive expression of Nrf2 was the corresponding fluorescein-labeled red light



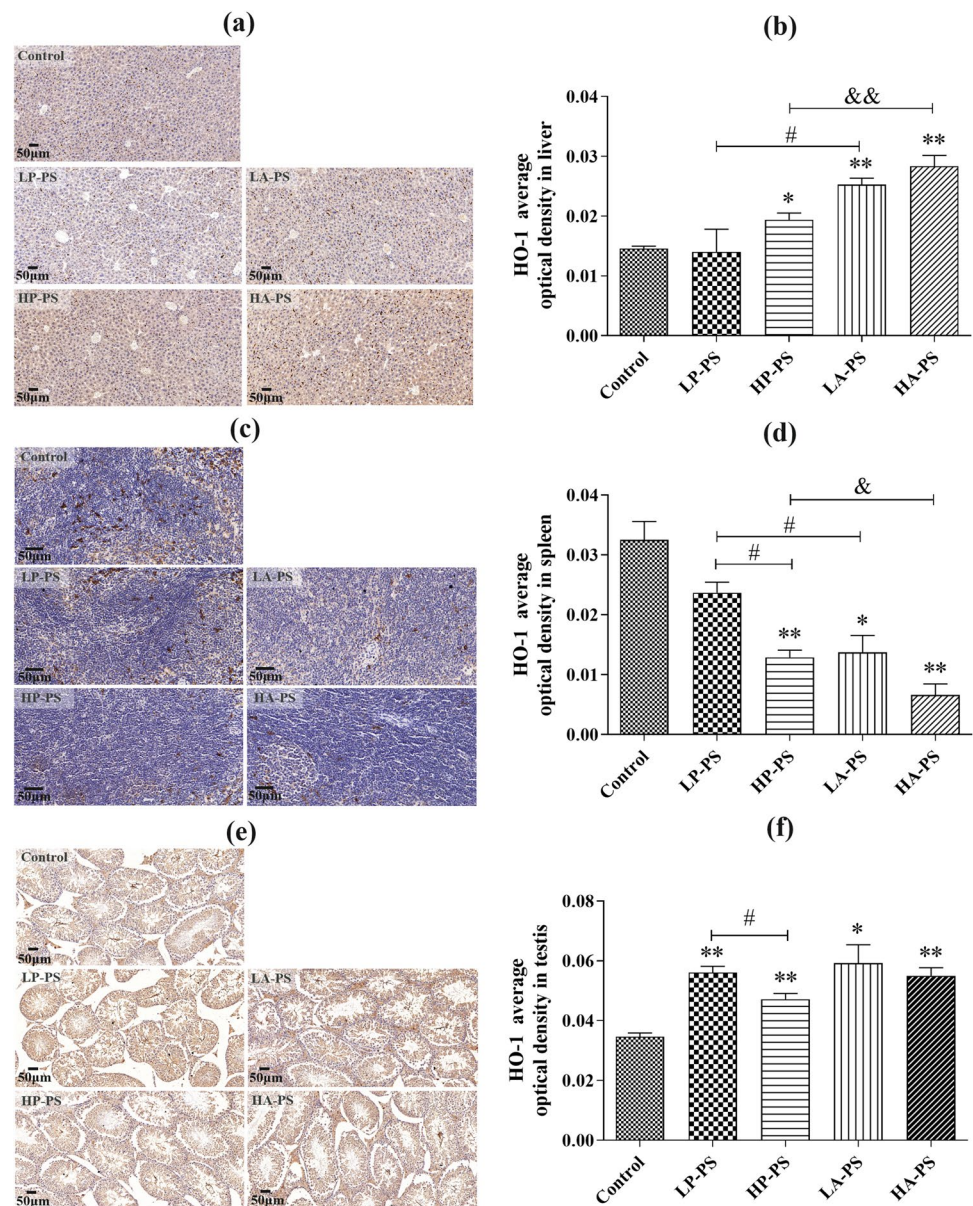
Nrf2 expression levels in the spleen induced by PS and APS exposure were significantly decreased compared to those in the control group (Fig. 3e and f, $p < 0.05$). Jeong et al. (2017) also suggested that PS exposure suppressed the expression of Nrf2 in *Paracyclopsina nana*. In this study, PS and APS inhibited Nrf2/HO-1 expression in the spleen and promoted IgA, demonstrating that the mouse antioxidant mechanism is closely related to the immune system. Exposure to LA-PS decreased the expression of HO-1 compared with that in the LP-PS group ($p < 0.01$), which implies that acute exposure to APS can induce a greater immune response in mice than PS. Reactive carbonyl compounds are closely related to the oxidative damage caused by abnormal Nrf2 activity, and can also cross link with proteins, resulting in physiological

disorders (Nishimoto et al. 2017; Ramasamy et al. 2006; Yin 1995). These results suggest that aged MPs could induce more serious oxidative damage in mouse organs than pristine MPs. Nrf2/HO-1 pathway could be used as a biomarker to assess the health risks of MPs under different environmental conditions.

PS and APS exposure damaged mouse liver structure and function

The liver is an important metabolic organ in humans, and its structure is closely related to its function (Heullitt 2013). Figure 5A shows that more severe histopathological changes in the liver, such as inflammatory infiltration, were

Fig. 4 Effects of PS and APS exposure on HO-1 levels in mouse organs. **a** Immunohistochemistry to detect HO-1 level in liver. **b** HO-1 average optical density in liver. **c** Immunohistochemistry to detect HO-1 level in spleen. **d** HO-1 average optical density in spleen. **e** Immunohistochemistry to detect HO-1 level in testis. **f** HO-1 average optical density in testis. * $p < 0.05$, ** $p < 0.01$, compared with the control group. # $p < 0.05$, compared with the LP-PS group. & $p < 0.05$, && $p < 0.01$, compared with the HP-PS group. For **a**, **c**, and **e**, the nucleus stained by hematoxylin was blue, and the positive expression of diaminobenzidine was brown



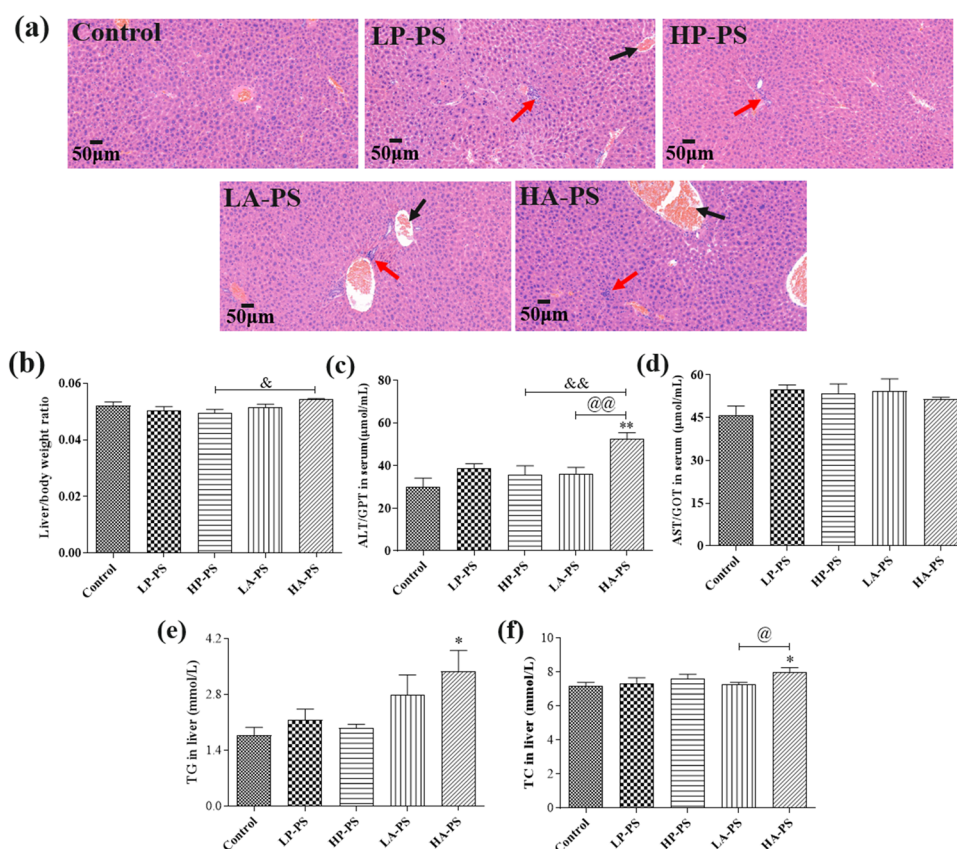
observed in the PS groups than in the control group. In addition, there were more erythrocytes and protein fluid in the portal vein lumen, showing that serious histopathological damage occurred in the LA-PS and HA-PS groups. Compared with HP-PS group, there were significant increases of liver and body weight (Fig. 5b, $p < 0.05$). ALT/GPT and AST/GOT levels in serum can reflect liver injuries and PS can induce metabolism disorders in mouse liver (Li et al. 2016; Huang et al. 2006; Lu et al. 2018). ALT/GPT levels in the HA-PS group were higher than those in the control and HP-PS groups (Fig. 5c, $p < 0.01$). There was an increase in AST/GOT levels in PS and APS groups compared with the control group (Fig. 5d). Previous studies have shown that PS exposure can also increase ALT/GPT levels, resulting in lipid metabolism disorders (Yu et al. 2018; Lu et al. 2018).

This suggests that ALT/GPT may be useful as an indicator of MP exposure, and APS could cause more serious liver damage and metabolism disorders.

Moreover, the liver plays an important role in lipid digestion and absorption (Sahini and Borlak 2016). Deng et al. (2020) found that PS-MPs exposure disturbs energy and lipid metabolism, inducing TG and TC accumulation in the liver. In this study, TG and TC levels increased in the LA-PS and HA-PS groups compared with the control (Fig. 5e and f). The accumulation of TG and TC in the liver suggests that APS exposure induced liver lipid metabolism disorder in mice ($p < 0.05$). In summary, APS caused greater structural and functional damage to mouse liver by inducing lipid metabolism disorders. An in vitro model system shows that PS-MPs adsorb lipase and form large lipid-MPs

Fig. 5 Effects of PS and APS exposure on the livers of mice. **a** Liver histopathology. **b** Liver/body weight ratio. **c** ALT/GPT levels in serum. **d** AST/GOT levels in serum. **e** TG levels in liver. **f** TC levels in liver.

* $p < 0.01$, ** $p < 0.01$, compared with the control group. & $p < 0.01$, && $p < 0.01$, compared with the HP-PS group. @ $p < 0.05$, @@ $p < 0.01$, compared with the LA-PS group. Solid arrow (black): increased erythrocyte and protein solution; solid arrow (red): inflammatory cell infiltration



heteroaggregates due to the high hydrophobicity of MPs, disrupting lipid metabolism and inducing hepatotoxicity (Tan et al. 2020; Cheng et al. 2022). Therefore, compared with PS, the APS with increased specific surface area may adsorb more lipid-related substances and interfere with the lipid metabolism process of the liver.

PS and APS exposure caused structural damage to the mouse gut

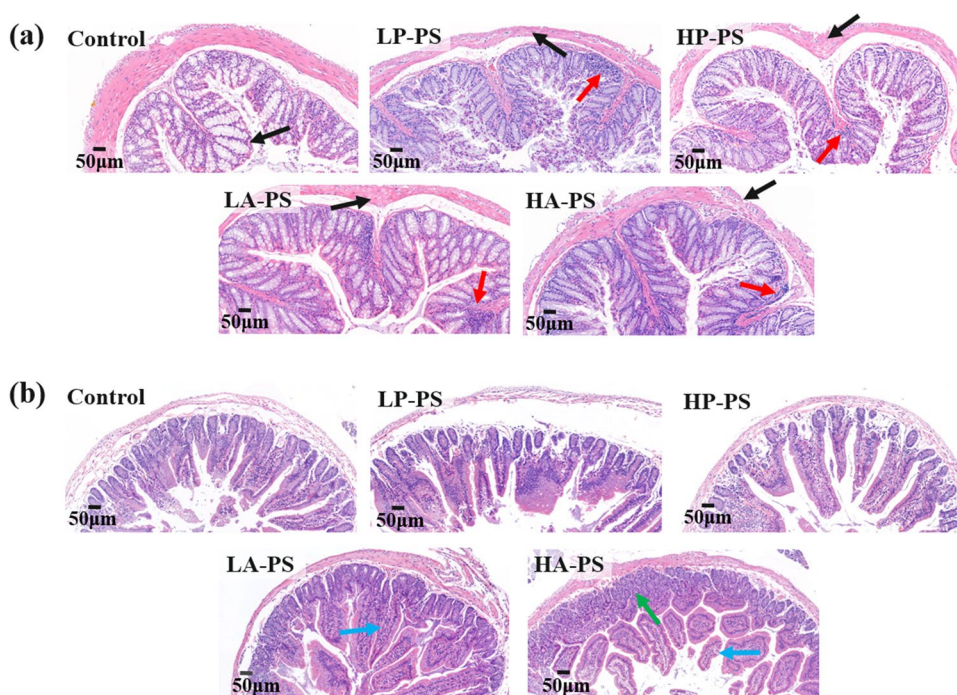
The gut and liver interact with each other, and liver damage causes gut dysfunction and, conversely, gut damage affects liver function (Zhang et al. 2021). In this study, pathological changes of the colon indicate that week-long exposure to LP-PS and HP-PS caused the reduction of mucosal myocytes and the increases of inflammatory cell infiltration compared with the control group (Fig. 6a). Simultaneously, pathological results from the ileum showed abnormal structural arrangements, such as the lamina propria cell proliferation, epithelial cell shedding, and disorders (Fig. 6b). Such gut damage has also been discovered in previous studies on PS (Lu et al. 2018). In the APS-exposed groups, gut villi were swollen and congested, and lamina propria lymphocytes and plasma cells were infiltrated compared with the PS groups. It is worth mentioning that structural damage induced by LA-PS and HA-PS exposure, such as swelling of colon and

ileum cells, was particularly significant. Compared with the pristine MPs, aged MP exposure can produce more adverse effects on colon cells (Xu et al. 2022), which suggests that aged MP can induce more potential health risks of gut dysfunction than pristine MPs. Consistently, this study suggest that the aging process induced abnormal chemical properties and surface roughness of PS-MPs, which may promote the occurrence of more serious damage to the liver-gut axis. Therefore, the potential health risks to human metabolism induced by aged MPs in the environment cannot be ignored.

PS and APS exposure induced immune and reproductive toxicity in mice

The spleen is an important lymphoid organ that can produce immune substances such as immunoglobulins (Li et al. 2020a, b). IgA, a protein that reflects the body's immunity, plays an important role in the immune system (Tomasi 1992). In this study, exposure to LP-PS caused significantly loose spleen globules compared with those in the HP-PS group (Fig. 7b), suggesting that acute exposure to PS could induce immunotoxicity. IgA expression levels in the LP-PS group significantly increased compared with those in the control group (Fig. 7a, $p < 0.05$). These results are consistent with previous studies in which PS caused spleen functional damage, inducing immunotoxicity (Park et al. 2020;

Fig. 6 Effects of PS and APS exposure on the gut of mice. **a** Mouse colon. **b** Mouse ileum. Solid arrow (black): reduction of mucosal myocytes; solid arrow (red): inflammatory cell infiltration; solid arrow (green): lamina propria cell proliferation; solid arrow (blue): disorders of intestinal epithelial cells



Li et al. 2020a, b). In addition, pathological changes of the spleen were observed in the HA-PS group compared with the HP-PS group, and there was a dose–effect relationship in the spleen damage induced by APS. Compared with the HP-PS group, the IgA levels in the HA-PS exposure group significantly increased ($p < 0.05$), suggesting that aged MPs can produce more severe immunotoxicity due to their unique physicochemical properties. These results show that exposure to PS and APS induced immune response of mice, and IgA could be used as a biomarker for the harmful effects of aged MP exposure.

Similarly, Hu et al. (2021) found that PS-MPs disturb the immune balance and cause reproductive toxicity in pregnant mice. As shown in Fig. 7c, the interstitium of the seminiferous tubules in testis was loose, and the normal spermatogenic cells fell off and were disorderly arranged. Furthermore, there were gaps in the boundary membrane of the seminiferous tubules, showing hyaline-like degeneration in PS and APS groups. A previous study suggested that PS induces reproductive toxicity in mice through oxidative stress (Xie et al. 2020). In this study, structural damage to the testis was more serious in the HA-PS group than in the HP-PS group, suggesting that aged MPs could produce more serious reproductive toxicity.

Toxicity of PS and APS exposure to mice

In this study, acute exposure to PS and APS can induce gut, liver, spleen, and testis tissue injury in mice (Fig. 8). Study on biochemical effects of aged MPs on birds shows that

ingested aged MPs can cause oxidative stress in their brain, liver, gut and other tissues, and birds may be the vectors of aged MPs in the natural environment, suggesting that aged MPs spread more widely in the natural environment and affect various organisms in the food chain (Silva de Souza et al. 2022). Cytotoxicity study of aged MPs found that the aging process could promote the release of dissolved organic matter, and induce the occurrence of oxidative stress (Xu et al. 2022). In this study, for the first time, a new perspective is used for analysis of the differences in toxicity of pristine and aged MPs to mice. APS not only induced more oxidative damage through carbonyl groups, but also caused immune and liver-gut axis function damage in mice, as well as potential reproductive toxicity effects. However, the mechanism of how microplastics induce oxidative damage and other effects in mouse organs remains to be further analyzed.

Importantly, as previous studies already indicated that MPs could induce immune and oxidative damage by increased IgA and abnormal expression of Nrf2 (Jeong and Choi 2019; Li et al. 2020a, b), we found that IgA and Nrf2/HO-1 may be used as indicators in health risk assessments of aged MPs. Furthermore, the results showed that PS and APS exposure significantly inhibited the antioxidant mechanism of Nrf2/HO-1 in spleen, and activated the antioxidant mechanism of Nrf2/HO-1 in liver and testis, indicating that APS exposure induced immune function damage in mice and there was a characterization of mouse tissues. Chronic PS-MP exposure can impair learning and memory in mice by inducing oxidative stress, and long-term exposure to UV-aged PS-MPs leads to neurotoxicity

(a)

Group	IgA in serum (mg/mL)
Control	~0.9
LP-PS	~0.95*
HP-PS	~0.9
LA-PS	~0.9
HA-PS	~0.95*

(b)

(c)

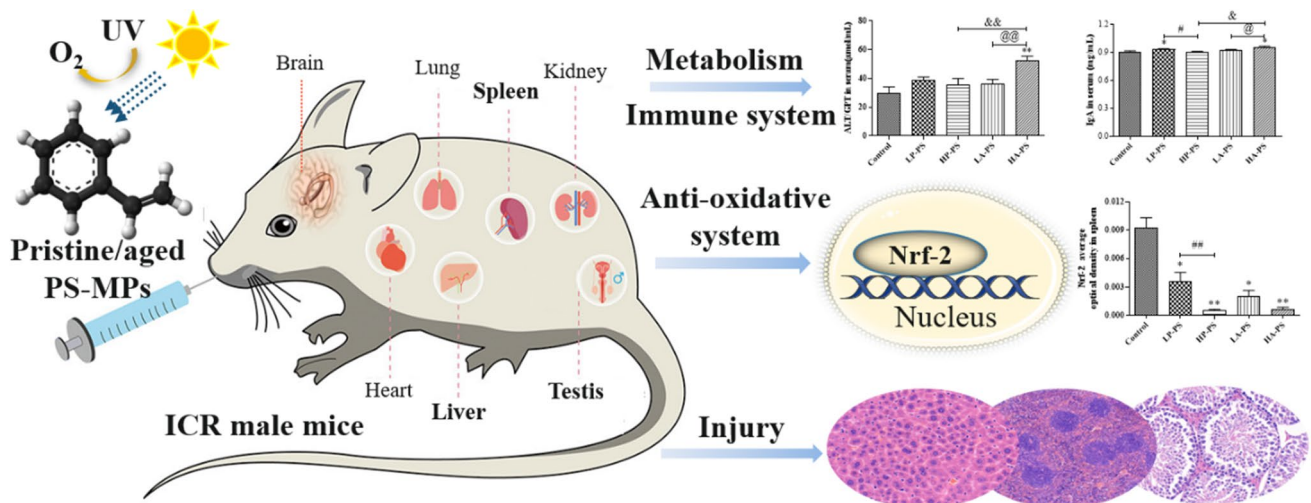


Fig. 8 Effects of pristine and aged PS exposure on the mouse organs

in nematode *Caenorhabditis elegans* (Wang et al. 2022; Chen et al. 2021). On contrary, this study found that acute exposure to PS and APS did not damage the brain tissue of mice (Fig. S3). Therefore, the toxic mechanism of aged MPs needs to be further analyzed, especially the chronic exposure to terrestrial organisms.

Conclusion

In summary, the present study showed that exposure to PS and APS caused antioxidant and structural damage to mouse organs, such as the gut, liver, spleen, and testis. The disorders of tissue-specific antioxidant system induced by abnormal expression of Nrf2/HO-1 involved in structural damage to the liver, spleen, and testis. PS and APS exposure significantly inhibited the Nrf2/HO-1 expression levels in spleen, and enhanced the Nrf2/HO-1 levels in liver and testis, and APS exposure induced more serious oxidative damage in mice. Detection of biochemical indicators in serum showed that HA-PS significantly harmed the liver and spleen function, indicating that APS potentially have potential toxic effects on the metabolism and immunity functions of mice than PS. These findings provide new insights regarding the potential risk of aged MPs in mice and play an important role in assessing the human health risk of MPs.

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Data availability All data generated or analyzed during this study are included in this published article.

Declarations

Ethical approval All animal experiment was approved by the Institutional Animal Care and Use Committee of Nanjing University.

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Consent to publish Not applicable.

Conflict of interest The authors declare no competing interests.

References

- Ayıkgoz Y, Salih Aydın M, Kankılıç N, Temiz E (2021) Nuclear factor erythroid 2-related factor 2 (Nrf2), tumor necrosis factor alpha protein (TNF- α), heme oxygenase-1 (HO-1) gene expressions during cardiopulmonary bypass. *Gene* 790:145690. <https://doi.org/10.1016/j.gene.2021.145690>
- Chen H, Hua X, Yang Y, Wang C, Jin L, Dong C, Chang Z, Ding P, Xiang M, Li H, Yu Y (2021) Chronic exposure to UV-aged microplastics induces neurotoxicity by affecting dopamine glutamate and serotonin neurotransmission in *Caenorhabditis elegans*. *J Hazard Mater* 419:126482. <https://doi.org/10.1016/j.jhazmat.2021.126482>
- Cheng W, Li X, Zhou Y, Yu H, Xie Y, Guo H, Wang H, Li Y, Feng Y, Wang Y (2022) Polystyrene microplastics induce hepatotoxicity and disrupt lipid metabolism in the liver organoids. *Sci Total Environ* 806(1):150328. <https://doi.org/10.1016/j.scitotenv.2022.150328>
- Cox KD, Covernton GA, Davies HL, Dower JF, Juanes F, Dudas SE (2019) Human consumption of microplastics. *Environ Sci Technol* 53(12):7068–7074. <https://doi.org/10.1021/acs.est.9b01517>
- Dela-Torre GE (2020) Microplastics: an emerging threat to food security and human health. *J Food Sci Technol* 57(5):1601–1608. <https://doi.org/10.1007/s13197-019-04138-1>
- Deng Y, Zhang Y, Lemos B, Ren H (2017) Tissue accumulation of microplastics in mice and biomarker responses suggest widespread health risks of exposure. *Sci Rep* 7:46687. <https://doi.org/10.1038/srep46687>
- Deng Y, Yan Z, Shen R, Wang M, Huang Y, Ren H, Zhang Y, Lemos B (2020) Microplastics release phthalate esters and cause aggravated adverse effects in the mouse gut. *Environ Int* 143:105916. <https://doi.org/10.1016/j.envint.2020.105916>
- Espinosa C, García Beltrán JM, Esteban MA, Cuesta A (2018) *In vitro* effects of virgin microplastics on fish head-kidney leucocyte activities. *Environ Pollut* 235:30–38. <https://doi.org/10.1016/j.envpol.2018.05.011>
- Fu D, Zhang Q, Fan Z, Qi H, Wang Z, Peng L (2019) Aged microplastics polyvinyl chloride interact with copper and cause oxidative stress towards *microalgae chlorella vulgaris*. *Aquat Toxicol* 216:105319. <https://doi.org/10.1016/j.aquatox.2019.105319>
- Heulitt MJ (2013) Structure determines function. *J Pediatr Intensive Care* 2(1):1–3. <https://doi.org/10.3233/PIC-13041>
- Hofker MH (2003) Introduction: the use of transgenic mice in biomedical research. *Methods Mol Biol* 209:1–8. <https://doi.org/10.1385/1-59259-340-2:01>
- Hou B, Wang F, Liu T, Wang Z (2021) Reproductive toxicity of polystyrene microplastics: *in vivo* experimental study on testicular toxicity in mice. *J Hazard Mater* 405:124028. <https://doi.org/10.1016/j.jhazmat.2021.124028>
- Hu M, Palić D (2020) Micro- and nano-plastics activation of oxidative and inflammatory adverse outcome pathways. *Redox Biol* 37:101620. <https://doi.org/10.1016/j.redox.2020.101620>
- Hu J, Qin X, Zhang J, Zhu Y, Zeng W, Lin Y, Liu X (2021) Polystyrene microplastics disturb maternal-fetal immune balance and cause reproductive toxicity in pregnant mice. *Reprod Toxicol* 106:42–50. <https://doi.org/10.1016/j.reprotox.2021.106422>
- Huang XJ, Choi YK, Im HS, Yarimaga O, Yoon E (2006) Aspartate aminotransferase (ast/got) and alanine aminotransferase (alt/gpt) detection techniques. *Sensors (basel)* 6(7):756–782. <https://doi.org/10.3390/s6070756>
- Jahnke A, Arp HHP, Escher BI, Gewert B, Gorokhova E, Kühnel D, Ogonowski M, Potthoff A, Rummel C, Schmitt-Jansen M, Toorman E, Matthew M (2017) Reducing uncertainty and confronting ignorance about the possible impacts of weathering plastic in the marine environment. *Environ Sci Technol Lett* 4:85–90. <https://doi.org/10.1021/acs.estlett.7b00008>

- Jeong J, Choi J (2019) Adverse outcome pathways potentially related to hazard identification of microplastics based on toxicity mechanisms. *Chemosphere* 231:249–255. <https://doi.org/10.1016/j.chemosphere>
- Jeong CB, Kang HM, Lee MC, Kim DH, Han J, Hwang DS, Souissi S, Lee SJ, Shin KH, Park HG, Lee JS (2017) Adverse effects of microplastics and oxidative stress-induced MAPK/Nrf2 pathway-mediated defense mechanisms in the marine copepod *Paracyclopina nana*. *Sci Rep* 7:41323. <https://doi.org/10.1038/srep41323>
- Jiang B, Kauffman AE, Li L, McFee W, Cai B, Weinstein J, Lead JR, Chatterjee S, Scott GI, Xiao S (2020) Health impacts of environmental contamination of micro- and nanoplastics: a review. *Environ Health Prev Med* 25(1):29. <https://doi.org/10.1186/s12199-020-00870-9>
- Jin X, Jin G, Pan X, Jin Q, Li G (1992) Investigation on the aging of polystyrene by UV-Vis and FTIR spectroscopy. *China Synth Resin Plast* 9(2):33–36
- Jin Y, Lu L, Tu W, Luo T, Fu Z (2019) Impacts of polystyrene microplastic on the gut barrier, microbiota and metabolism of mice. *Sci Total Environ* 649:308–317. <https://doi.org/10.1016/j.scitotenv>
- Kim JS, Lee HJ, Kim SK, Kim HJ (2018) Global pattern of microplastics (MPs) in commercial food-grade salts: Sea salt as an indicator of seawater MP pollution. *Environ Sci Technol* 52(21):12819–12828. <https://doi.org/10.1021/acs.est.8b04180>
- Lambert S, Wagner M (2016) Characterisation of nanoplastics during the degradation of polystyrene. *Chemosphere* 145:265–268. <https://doi.org/10.1016/j.chemosphere>
- Leung MM, Ho YW, Lee CH, Wang Y, Hu M, Kwok KWH, Chua SL, Fang JK (2021) Improved Raman spectroscopy-based approach to assess microplastics in seafood. *Environ Pollut* 289:117648. <https://doi.org/10.1016/j.envpol>
- Li MX, Zhao H, Bi XY, Li ZY, Yao XS, Li H, Huang Z, Han Y, Zhou JG, Zhao JJ, Zhang YF, Zhao DB, Cai JQ (2016) Lactate dehydrogenase is a prognostic indicator in patients with hepatocellular carcinoma treated by sorafenib: results from the real life practice in HBV endemic area. *Oncotarget* 7(52):86630–86647. <https://doi.org/10.18632/oncotarget>
- Li B, Ding Y, Cheng X, Sheng D, Xu Z, Rong Q, Wu Y, Zhao H, Ji X, Zhang Y (2020a) Polyethylene microplastics affect the distribution of gut microbiota and inflammation development in mice. *Chemosphere* 244:125492. <https://doi.org/10.1016/j.chemosphere>
- Li W, Li L, Li W, Chopp M, Venkat P, Zacharek A, Chen ZL, Landschoot-Ward JL, Chen JL (2020b) Spleen associated immune-response mediates brain-heart interaction after intracerebral hemorrhage. *Exp Neurol* 327:113209. <https://doi.org/10.1016/j.expneurol>
- Liang Y, Yang X, Wang Y, Liu R, Mao L (2021) Influence of polystyrene microplastics on rotifer (*Brachionus calyciflorus*) growth, reproduction, and antioxidant responses. *Aquat Ecol* 55:1097–1111. <https://doi.org/10.1007/s10452-021-09885-y>
- Liu G, Zhu Z, Yang Y, Sun Y, Yu F, Ma J (2019a) Sorption behavior and mechanism of hydrophilic organic chemicals to virgin and aged microplastics in freshwater and seawater. *Environ Pollut* 246:26–33. <https://doi.org/10.1016/j.envpol>
- Liu P, Qian L, Wang H, Zhan X, Lu K, Gu C, Gao S (2019b) New insights into the aging behavior of microplastics accelerated by advanced oxidation processes. *Environ Sci Technol* 53(7):3579–3588. <https://doi.org/10.1021/acs.est.9b00493>
- Lu Y, Zhang Y, Deng Y, Jiang W, Zhao Y, Geng J, Ding L, Ren H (2016) Uptake and accumulation of polystyrene microplastics in zebrafish (*Danio rerio*) and toxic effects in liver. *Environ Sci Technol* 50(7):4054–4060. <https://doi.org/10.1021/acs.est.6b00183>
- Lu L, Wan Z, Luo T, Fu Z, Jin Y (2018) Polystyrene microplastics induce gut microbiota dysbiosis and hepatic lipid metabolism disorder in mice. *Sci Total Environ* 631–632:449–458. <https://doi.org/10.1016/j.scitotenv>
- Moi P, Chan K, Asunis I, Cao A, Kan YW (1994) Isolation of NF-E2-related factor 2 (Nrf2), a NF-E2-like basic leucine zipper transcriptional activator that binds to the tandem NF-E2/AP1 repeat of the beta-globin locus control region. *Proc Natl Acad Sci U S A* 91(21):9926–9930. <https://doi.org/10.1073/pnas.91.21.9926>
- Nicole W (2021) Microplastics in seafood: How much are people eating? *Environ Health Perspect* 129(3):34001. <https://doi.org/10.1289/EHP8936>
- Nishimoto S, Koike S, Inoue N, Suzuki T, Ogasawara Y (2017) Activation of Nrf2 attenuates carbonyl stress induced by methylglyoxal in human neuroblastoma cells: Increase in GSH levels is a critical event for the detoxification mechanism. *Biochem Biophys Res Commun* 483(2):874–879. <https://doi.org/10.1016/j.bbrc.2017.01.024>
- Niture SK, Khatri R, Jaiswal AK (2014) Regulation of Nrf2—an update. *Free Radic Biol Med* 66:36–44. <https://doi.org/10.1016/j.freeradbiomed>
- Oliveri Conti G, Ferrante M, Banni M, Favara C, Nicolosi I, Cristaldi A, Fiore M, Zuccarello P (2020) Micro- and nano-plastics in edible fruit and vegetables. The first diet risks assessment for the general population. *Environ Res* 187:109677. <https://doi.org/10.1016/j.envres>
- Park EJ, Han JS, Park EJ, Seong E, Lee GH, Kim DW, Son HY, Han HY, Lee BS (2020) Repeated-oral dose toxicity of polyethylene microplastics and the possible implications on reproduction and development of the next generation. *Toxicol Lett* 324:75–85. <https://doi.org/10.1016/j.toxlet>
- Pigeolet E, Corbisier P, Houbion A, Lambert D, Remacle J (1990) Glutathione peroxidase, superoxide dismutase, and catalase inactivation by peroxides and oxygen derived free radicals. *Mech Ageing Dev* 51(3):283–297. [https://doi.org/10.1016/0047-6374\(90\)90078-T](https://doi.org/10.1016/0047-6374(90)90078-T)
- Piñeiro-Hermida S, Martínez P, Bosso G, Flores JM, Saraswati S, Connor J, Lemaire R, Blasco MA (2022) Consequences of telomere dysfunction in fibroblasts, club and basal cells for lung fibrosis development. *Nat Commun* 13(1):5656. <https://doi.org/10.1038/s41467-022-32771-6>
- Ragusa A, Svelato A, Santacroce C, Catalano P, Notarstefano V, Carnevali O, Papa F, Rongioletti MCA, Baiocco F, Draghi S, D'Amore E, Rinaldo D, Matta M, Giorgini E (2021) Plasticenta: first evidence of microplastics in human placenta. *Environ Int* 146:106274. <https://doi.org/10.1016/j.envint.2020.106274>
- Ramasamy R, Yan SF, Schmidt AM (2006) Methylglyoxal comes of AGE. *Cell* 124(2):258–260. <https://doi.org/10.1016/j.cell.2006.01.002>
- Sahini N, Borlak J (2016) Genomics of human fatty liver disease reveal mechanistically linked lipid droplet-associated gene regulations in bland steatosis and nonalcoholic steatohepatitis. *Transl Res* 177:41–69. <https://doi.org/10.1016/j.trsl>
- Schultz TW, Netzeva TI, Roberts DW, Cronin MT (2005) Structure-toxicity relationships for the effects to tetrahymena pyriformis of aliphatic, carbonyl-containing, alpha, beta-unsaturated chemicals. *Chem Res Toxicol* 18(2):330–341. <https://doi.org/10.1021/tx049833j>
- Shruti VC, Pérez-Guevara F, Kutralam-Muniasamy G (2020) Metro station free drinking water fountain- a potential “microplastics hotspot” for human consumption. *Environ Pollut* 261:114227. <https://doi.org/10.1016/j.envpol>
- Silva de Souza S, Nascimento Freitas Í, de Oliveira Gonçalves S, Marinho da Luz T, Pereira da Costa Araújo A, Rajagopal R, Balasubramani G, Rahman MM, Malafaia G (2022) Toxicity induced via ingestion of naturally-aged polystyrene microplastics by a small-sized terrestrial bird and its potential role as vectors for the

- dispersion of these pollutants. *J Hazard Mater* 128814 <https://doi.org/10.1016/j.jhazmat.2022.128814>
- Song X, Lin Q, Chen C, Chen J, Hu C (2018) Identification of original and recycled polystyrene containers by Fourier transform infrared spectroscopy and principal component analysis. *Packag Technol Sci* 6:70–70. <https://doi.org/10.1002/pts.2378>
- Suman TY, Jia PP, Li WG, Junaid M, Xin GY, Wang Y, Pei DS (2020) Acute and chronic effects of polystyrene microplastics on brine shrimp: First evidence highlighting the molecular mechanism through transcriptome analysis. *J Hazard Mater* 400:123220. <https://doi.org/10.1016/j.jhazmat.2020.123220>
- Sun XD, Yuan XZ, Jia Y, Feng LJ, Zhu FP, Dong SS, Liu J, Kong X, Tian H, Duan JL, Ding Z, Wang SG, Xing B (2020) Differentially charged nanoplastics demonstrate distinct accumulation in *Arabidopsis thaliana*. *Nat Nanotechnol* 15(9):755–760. <https://doi.org/10.1038/s41565-020-0707-4>
- Suzuki K, Taniguchi N, Ookawara T, Ohno H (1999) Superoxide dismutase (SOD). *Nihon Rinsho* 57:499–502. <https://doi.org/10.1038/scibx.2008.45>
- Tan H, Yue T, Xu Y, Zhao J, Xing B (2020) Microplastics reduce lipid digestion in simulated human gastrointestinal system. *Environ Sci Technol* 54(19):12285–12294. <https://doi.org/10.1021/acs.est.0c02608>
- Tang W, Jiang YF, Ponnusamy M, Diallo M (2014) Role of Nrf2 in chronic liver disease. *World J Gastroenterol* 20(36):13079–13087. <https://doi.org/10.3748/wjg.v20.i36.13079>
- Tomasi TB (1992) The discovery of secretory IgA and the mucosal immune system. *Immunol Today* 13(10):416–418. [https://doi.org/10.1016/0167-5699\(92\)90093-M](https://doi.org/10.1016/0167-5699(92)90093-M)
- Vroom RJE, Koelmans AA, Besseling E, Halsband C (2017) Aging of microplastics promotes their ingestion by marine zooplankton. *Environ Pollut* 231(1):987–996. <https://doi.org/10.1016/j.envpol.2017.09.087>
- Walczak AP, Hendriksen PJ, Woutersen RA, van der Zande M, Undas AK, Helsdingen R, van den Berg HH, Rietjens IM, Bouwmeester H (2015a) Bioavailability and biodistribution of differently charged polystyrene nanoparticles upon oral exposure in rats. *J Nanopart Res* 17(5):231. <https://doi.org/10.1007/s11051-015-3029-y>
- Walczak AP, Kramer E, Hendriksen PJ, Tromp P, Helsper JP, van der Zande M, Rietjens IM, Bouwmeester H (2015b) Translocation of differently sized and charged polystyrene nanoparticles in *in vitro* intestinal cell models of increasing complexity. *Nanotoxicol* 9(4):453–461. <https://doi.org/10.3109/17435390.2015.1035390>
- Wan Z, Wang C, Zhou J, Shen M, Wang X, Fu Z, Jin Y (2019) Effects of polystyrene microplastics on the composition of the microbiome and metabolism in larval zebrafish. *Chemosphere* 217:646–658. <https://doi.org/10.1016/j.chemosphere.2019.03.088>
- Wang T, Ma Y, Ji R (2021) Aging processes of polyethylene mulch films and preparation of microplastics with environmental characteristics. *Bull Environ Contam Toxicol* 107(4):736–740. <https://doi.org/10.1007/s00128-020-02975-x>
- Wang S, Han Q, Wei Z, Wang Y, Xie J, Chen M (2022) Polystyrene microplastics affect learning and memory in mice by inducing oxidative stress and decreasing the level of acetylcholine. *Food Chem Toxicol* 162:112904. <https://doi.org/10.1016/j.fct.2022.112904>
- Wong SL, Nyakuma BB, Wong KY, Lee CT, Lee TH (2020) Lee CH (2009–2019) Microplastics and nanoplastics in global food webs: a bibliometric analysis. *Mar Pollut Bull* 158:111432. <https://doi.org/10.1016/j.marpolbul.2020.111432>
- Wu B, Wu X, Liu S, Wang Z, Chen L (2019) Size-dependent effects of polystyrene microplastics on cytotoxicity and efflux pump inhibition in human Caco-2 cells. *Chemosphere* 221:333–341. <https://doi.org/10.1016/j.chemosphere.2019.03.088>
- Xie X, Deng T, Duan J, Xie J, Yuan J, Chen M (2020) Exposure to polystyrene microplastics causes reproductive toxicity through oxidative stress and activation of the p38 MAPK signaling pathway. *Ecotoxicol Environ Saf* 190:110133. <https://doi.org/10.1016/j.ecoenv.2020.110133>
- Xu Y, Huang D, Liu P, Ouyang Z, Jia H, Guo X (2022) The characteristics of dissolved organic matter release from UV-aged microplastics and its cytotoxicity on human colonic adenocarcinoma cells. *Sci Total Environ* 826:154177. <https://doi.org/10.1016/j.scitotenv.2022.154177>
- Yang H, Lai H, Huang J, Sun L, Mennigen JA, Wang Q, Liu Y, Jin Y, Tu W (2020) Polystyrene microplastics decrease F-53B bioaccumulation but induce inflammatory stress in larval zebrafish. *Chemosphere* 255:127040. <https://doi.org/10.1016/j.chemosphere.2020.127040>
- Yin D (1995) Studies on age pigments evolving into a new theory of biological aging. *Gerontol* 2:159–172. <https://doi.org/10.1159/000213740>
- Yu P, Liu Z, Wu D, Chen M, Lv W, Zhao Y (2018) Accumulation of polystyrene microplastics in juvenile *Eriocheir sinensis* and oxidative stress effects in the liver. *Aquat Toxicol* 200:28–36. <https://doi.org/10.1016/j.aquatox.2018.08.008>
- Zhang Q, Xu EG, Li J, Chen Q, Ma L, Zeng EY, Shi H (2020) A review of microplastics in table salt, drinking water, and air: direct human exposure. *Environ Sci Technol* 54(7):3740–3751. <https://doi.org/10.1021/acs.est.9b04535>
- Zhang H, Liang Y, Wu P, Shi X, Zhang G, Cai Z (2021) Continuous dermal exposure to triclocarban perturbs the homeostasis of liver-gut axis in mice: insights from metabolic interactions and microbiome shifts. *Environ Sci Technol* 55(8):5117–5127. <https://doi.org/10.1021/acs.est.0c08273>

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